

WORKSHOP · SESSION 1

AI 101

What it is, how it works, and how to use it well

Teach · Explore · Demystify · Try it

START HERE

What we mean by “AI”

Artificial intelligence — computers doing tasks we link with human thinking

Machine learning — learns patterns from examples, not hand-written rules

Generative AI — creates new text, images, audio or code

Large language models — e.g. ChatGPT, Claude, Gemini

NOT AS NEW AS IT LOOKS

Seventy years in six moments

1956



“Artificial intelligence”
named at
Dartmouth

1997



Deep Blue
beats chess
champion
Kasparov

2012



AlexNet sparks
the
deep-learning
boom

2016



AlphaGo beats
Go champion
Lee Sedol

2017



The
Transformer
architecture is
published

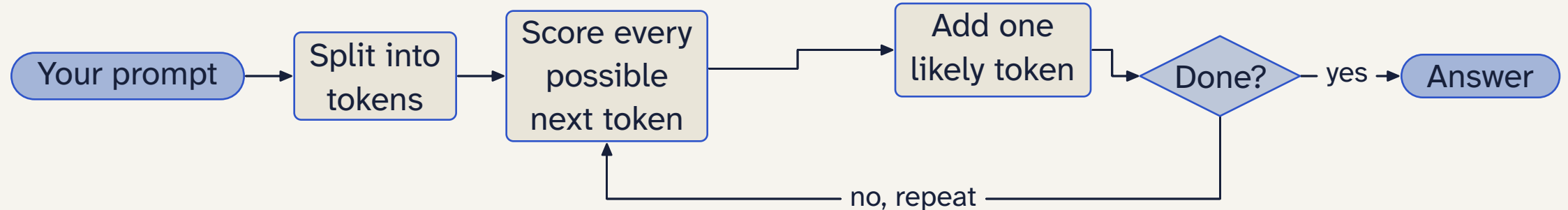
2022



ChatGPT
brings
chatbots to
everyone

UNDER THE HOOD

How a language model answers



A **token** is a chunk of text, often part of a word. The loop runs once per token, so a long answer is thousands of small predictions.

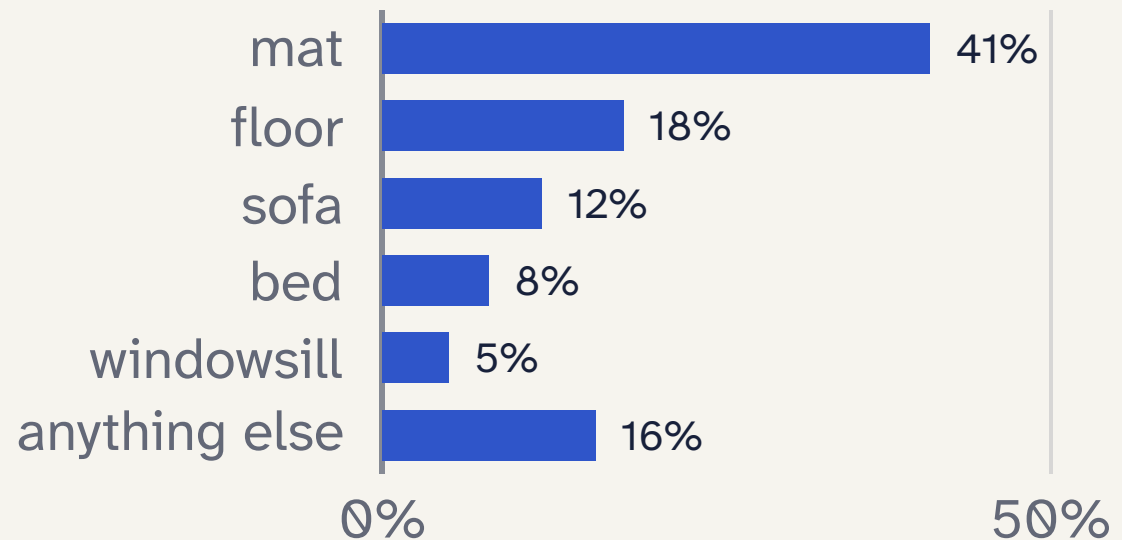
ONE STEP, UP CLOSE

“The cat sat on the ...”

Every candidate word gets a probability. The model then picks one: usually a likely word, not always the top one.

Illustrative numbers for teaching, not output from a real model.

Next-word probability (illustrative)



Strengths and limits

Good at

- Drafting and rewriting
- Summarising long documents
- Explaining at your level
- Brainstorming options
- Writing and fixing code

Watch out for

- Confident mistakes
- Out-of-date knowledge
- Arithmetic without tools
- Bias learned from data
- Facts it cannot check

“

We can only see a short distance ahead, but we can see plenty there that needs to be done.

Alan Turing, “Computing Machinery and Intelligence”, 1950

Four habits of a good prompt

1

Give context

Who you are, who it is for, what you know.

2

Be specific

Length, format and tone: “5 bullets for a manager”.

3

Show an example

Paste a sample of the style you want.

4

Iterate

Treat the first answer as a draft.

YOUR TURN

Use it well, starting this week

Check facts, numbers and quotes.

Protect personal and confidential data.

Say so when AI helped with shared work.

Stay in charge: you own the result.

Try this week

1. Pick one low-risk task.
2. Ask three different ways.
3. Note what worked.